

# Use-Case Flow Specification

Core Use Case: Check Bus ETA

## 1. Use Case Name

Check Bus ETA

## 2. Primary Actor

Commuter

## 3. Preconditions

1. The system is operational.
2. Bus GPS data is available.
3. The commuter has selected a bus stop or route.

## 4. Postconditions

1. The commuter receives the estimated arrival time.
2. The displayed ETA reflects the latest available GPS information.

## 5. Main Success Scenario

1. Commuter opens the bus tracking system.
2. Commuter selects a route or bus stop.
3. System retrieves the latest GPS data.
4. System determines the bus's current position.
5. System calculates the estimated arrival time.
6. System displays the ETA to the commuter.
7. System updates the ETA when new GPS data is received.

## 6. Alternate Flow

**A1 – Live GPS data unavailable:** If live GPS data is temporarily unavailable, the system displays the most recent available ETA and informs the commuter that live tracking data is temporarily unavailable.

**Relationship to system requirements:** This use case depends on FR-001 (GPS ingestion and ETA calculation) and supports FR-003 (display ETA to commuters).